12. The Analysis of Variance

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SS26 MTH344: Mathematical Statistics

ULSAN NATIONAL INSTITUTE OF SCIENCE AND TECHNOLOGY

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Lecture Topics

- In Chapter 11, we studied testing methods for comparing two distributions, such as placebo and treatment groups. In Chapter 12, we will expand these methods to analyze data from multiple distributions, with a particular focus on the **An**alysis **o**f **Va**riance (ANOVA).
- See Chapter 12 in **Mathematical Statistics and Data Analysis [MSDA]** (Rice, 2007) for details. Selected examples and text excerpts are reproduced from [MSDA] for educational use only under fair use.
- This course presents content mostly following the textbooks, but also provides substantial unique and advanced material.
- Most of nonparameteric topics are assigned as reading materials and will not be covered in this course.

Lecture Topics: The One-Way Layout

Group 1 (Placebo)	Group 2 (Treatment)
Y_{11}	Y_{21}
$\vdots$	$\vdots$
Y_{n_1}	$\vdots$
	Y_{2n_2}

- In Chapter 11, we denoted the data from two populations as $X_1, ..., X_n$ and $Y_1, \ldots, Y_m$.
- To accommodate multiple populations, we now represent them as $Y_{11}, \ldots, Y_{1n_1}$ and $Y_{21}, \ldots, Y_{2n_2}$.

Lecture Topics: The One-Way Layout

Group 1	Group 2	$\cdots$	Group I
Y_{11}	Y_{21}	$\cdots$	Y_{I1}
$\vdots$	$\vdots$		$\vdots$
Y_{1n_1}	$\vdots$		$\vdots$
	$\vdots$		Y_{In_I}
	Y_{2n_2}		

- The *one-way layout* refers to an experimental design where data are classified according to a single categorical factor (e.g., treatment type).
- When we have I groups under treatment $1, 2, \ldots, I$, we denote outcomes by Y_{ij}, where $i = 1, \ldots, I$ and $j = 1, \ldots, n_i$.
- We are usually interested in testing:

$$H_0 : \text{There is no treatment effect} \quad \text{vs.} \quad H_1 : \text{Not } H_0, \tag{1}$$

Lecture Topics: The One-Way Layout

Lab 1	Lab 2	Lab 3	Lab 4	Lab 5	Lab 6	Lab 7
4.13	3.86	4.00	3.88	4.02	4.02	4.00
4.07	3.85	4.02	3.88	3.95	3.86	4.02
4.04	4.08	4.01	3.91	4.02	3.96	4.03
4.07	4.11	4.01	3.95	3.89	3.97	4.04
4.05	4.08	4.04	3.92	3.91	4.00	4.10
4.04	4.01	3.99	3.97	4.01	3.82	3.81
4.02	4.02	4.03	3.92	3.89	3.98	3.91
4.06	4.04	3.97	3.90	3.89	3.99	3.96
4.10	3.97	3.98	3.97	3.99	4.02	4.05
4.04	3.95	3.98	3.90	4.00	3.93	4.06

Tab. 12.1. [MSDA] Example data when $I = 7$ and $n_i = 10$ for all i.

Lecture Topics: The One-Way Layout

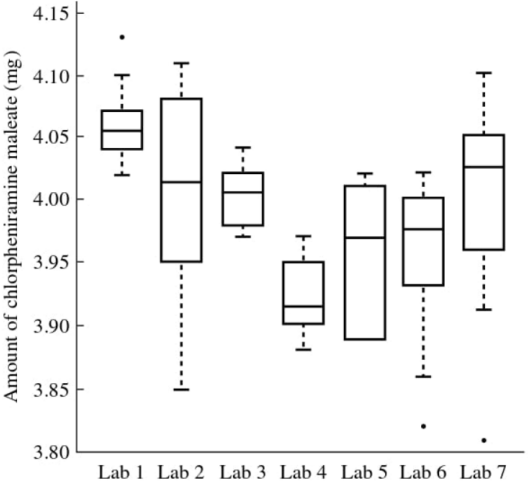


Fig. 12.1. [MSDA] Boxplots of data in Table 12.1.

Lecture Topics: The Two-Way Layout

	Factor B		
Factor A	Level 1	$\cdots$	Level J
Level 1	$Y_{111}, \ldots, Y_{11n_{11}}$	$\cdots$	$Y_{1J1}, \ldots, Y_{1Jn_{1J}}$
$\vdots$	$\vdots$	$\ddots$	$\vdots$
Level I	$Y_{I11}, \ldots, Y_{I1n_{I1}}$	$\cdots$	$Y_{IJ1}, \ldots, Y_{IJn_{IJ}}$

- While the one-way layout addresses the effect of a single factor, we are often interested in the effects of multiple factors and their interactions.
- The *two-way layout* refers to an experimental design where data are classified according to two categorical factors (e.g., Factor A and Factor B).
- For two factors with I and J levels, respectively, and n_{ij} observations per combination, we denote the outcomes by Y_{ijk}, where $i = 1, \ldots, I$, $j = 1, \ldots, J$, and $k = 1, \ldots, n_{ij}$.

Lecture Topics: The Two-Way Layout

	Factor B		
Factor A	Level 1	$\cdots$	Level J
Level 1	$Y_{111}, \ldots, Y_{11n_{11}}$	$\cdots$	$Y_{1J1}, \ldots, Y_{1Jn_{1J}}$
$\vdots$	$\vdots$	$\ddots$	$\vdots$
Level I	$Y_{I11}, \ldots, Y_{I1n_{I1}}$	$\cdots$	$Y_{IJ1}, \ldots, Y_{IJn_{IJ}}$

- We are usually interested in testing main effects and their interaction:

$$\begin{aligned} H_0^A &: \text{There is no main effect of Factor A} \\ H_0^B &: \text{There is no main effect of Factor B} \\ H_0^{AB} &: \text{There is no interaction effect between A and B} \end{aligned} \tag{2}$$

Lecture Topics: The Two-Way Layout

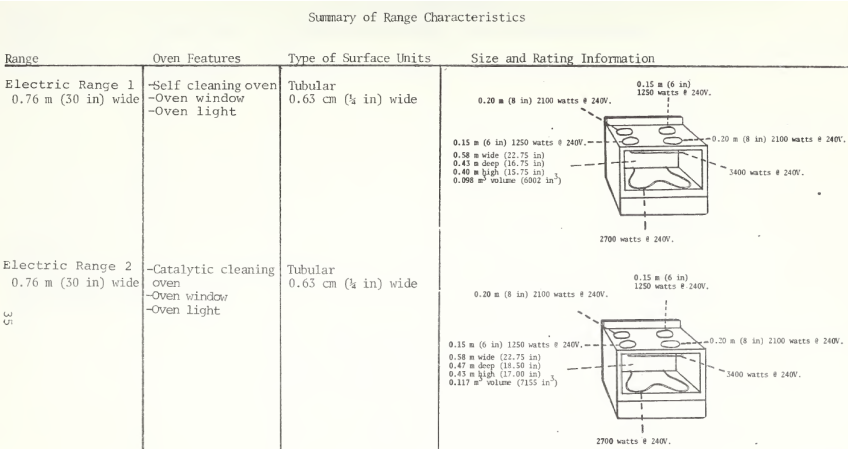
- As a case study, Fechter and Porter (1979) analyzed the energy consumption of appliances in response to the need for research supporting the national appliance energy conservation program.
- They measured the amount of energy consumed by menu days (Factor A) and kitchen ranges (Factor B). A total of fifty-eight non-professional cooks prepared the food.

Lecture Topics: The Two-Way Layout

	MENU DAY #1	MENU DAY #2	MENU DAY #3	MENU DAY #4	MENU DAY #5	MENU DAY #6	MENU DAY #7
BREAKFAST	Half Grapefruit Oatmeal with raisins, brown sugar Perked coffee Milk	Cranberry Juice Dry cereal with sliced bananas and milk Toast Butter or margarine Hot water (coffee, tea, chocolate) #4	Orange Juice Scrambled Eggs with bacon strips Toast - Jam Butter or margarine Hot water (coffee, tea, chocolate) Milk #7	Tomato Juice Dry Cereal with milk Toasted English Muffins Jam Butter or margarine Hot water (coffee, tea, chocolate) #10	Orange Juice Soft cooked eggs Toast - Jam Butter or margarine Hot water (coffee, tea, chocolate) Milk #13	Grapefruit Juice Dry cereal with milk Toast Butter or margarine Hot water (coffee, tea, chocolate) #16	Sliced bananas in Orange Juice Pancakes with syrup & sausages Butter or margarine Perked coffee Milk #19
MEAL NO.	#1						
LUNCH/ DINNER	Grilled cheese sandwiches Relish tray (carrot sticks, tomato, radishes, pickles) Fresh fruit Hot water (coffee, tea) Milk #2	Vegetable soup Cold beef sandwiches Pickles Hot water (coffee, tea) Milk #5	Hot dogs on buns Potato chips Pickles Fresh fruit Hot water (coffee, tea) Milk #8	Chicken noodle soup Tuna and egg salad sandwiches Cookies Hot water (coffee, tea) Milk #11	Leftover beef stew Fresh fruit salad Hot water (coffee, tea) Milk #14	Egg salad sandwich on toast Hot water (coffee, tea) Milk #17	Peanut butter & jelly sandwiches Carrot strips and celery sticks Leftover desserts or fruits Hot water (coffee, tea) Milk #20
MEAL NO.							
DINNER/ SUPPER	Beef pot roast with potatoes, carrots, onions, and gravy Tossed greens with radish slices French dressing Hot rolls Butter or margarine Apple crisp Coffee or tea Milk #3	Baked chicken with gravy Mashed potatoes Green beans Fruit-in-gelatin salad Refrigerated biscuits Butter or margarine Coffee or tea Milk #6	Beef stew (with carrots, peas, and potatoes) Tossed greens with French dressing Brown & Serve rolls Butter or margarine Vanilla pudding with strawberry halves in syrup Coffee or tea Milk #9	Spaghetti with meat sauce Grated parmesan cheese Lettuce wedge with Italian dressing Garlic bread Whipped gelatin Coffee or tea Milk #12	Braised pork chops Skillet scalloped potatoes Buttered spinach Red hot applesauce Bread Butter or margarine Lemon meringue pie Coffee or tea Milk #15	Broiled fish Rice pilaf Peas with pearl onions Broiled tomato Hard rolls Butter or margarine Tapioca pudding with fruit topping Coffee or tea Milk #18	Steak Baked potatoes Sauteed mushrooms Broccoli Tossed salad Hot French bread Butter or margarine Chocolate cake Coffee or tea Milk #21
MEAL NO.							

Table 1 in Fechter and Porter (1979) for menu days (Factor A).

Lecture Topics: The Two-Way Layout



Summary of Range Characteristics

Range	Oven Features	Type of Surface Units	Size and Rating Information
Electric Range 1 0.76 m (30 in) wide	-Self cleaning oven -Oven window -Oven light	Tubular 0.63 cm (¼ in) wide	0.15 m (6 in) 1250 watts @ 240V. 0.20 m (8 in) 2100 watts @ 240V. 0.15 m (6 in) 1250 watts @ 240V. 0.20 m (8 in) 2100 watts @ 240V. 0.58 m wide (22.75 in) 0.43 m deep (16.75 in) 0.40 m high (15.75 in) 0.098 m^3 volume (6002 in^3) 3400 watts @ 240V. 2700 watts @ 240V.
Electric Range 2 0.76 m (30 in) wide	-Catalytic cleaning oven -Oven window -Oven light	Tubular 0.63 cm (¼ in) wide	0.15 m (6 in) 1250 watts @ 240V. 0.20 m (8 in) 2100 watts @ 240V. 0.15 m (6 in) 1250 watts @ 240V. 0.20 m (8 in) 2100 watts @ 240V. 0.58 m wide (22.75 in) 0.47 m deep (18.50 in) 0.43 m high (17.00 in) 0.117 m^3 volume (7155 in^3) 3400 watts @ 240V. 2700 watts @ 240V.

A figure in Fechter and Porter (1979) for ranges (Factor B).

Lecture Topics: The Two-Way Layout

	Electric Ranges						Gas Ranges						
	E1	E2	E3	E4	E5	Mean	G1	G2	G3	G4	G5	Mean	
Menu Day	kWh	kWh	kWh	kWh	kWh	kWh	E-kWh	E-kWh	E-kWh	E-kWh	E-kWh	E-kWh	Menu Day
1	3.97	4.24	4.44	3.91	3.32	3.97	8.25	8.26	6.55	8.21	6.69	7.59	1
2	2.39	2.61	2.82	2.31	2.06	2.43	5.12	4.81	3.87	4.81	3.99	4.52	2
3	2.76	2.75	3.01	2.44	2.55	2.69	5.32	4.37	3.76	4.67	4.37	4.49	3
4	3.34	3.68	3.73	3.05	3.10	3.38	8.00	6.50	5.38	6.51	5.60	6.39	4
5	3.02	3.37	3.54	2.89	2.67	3.09	6.97	6.26	5.03	6.40	5.60	6.05	5
6	3.55	3.62	3.64	2.79	2.99	3.31	7.65	5.84	5.23	6.24	5.73	6.13	6
7	4.04	4.22	4.24	3.45	3.17	3.82	7.86	7.31	5.87	6.64	6.03	6.74	7
Mean of the Means	3.29	3.49	3.63	2.97*	2.83*	3.24	7.02	6.19	5.09	6.21*	5.43*	5.98	Mean of the Means

Table 3 in Fechter and Porter (1979). Summary of average energy consumptions by menu days (Factor A) and ranges (Factor B). For electric ranges, $I = 7$ and $J = 5$.

Lecture Topics: The Two-Way Layout

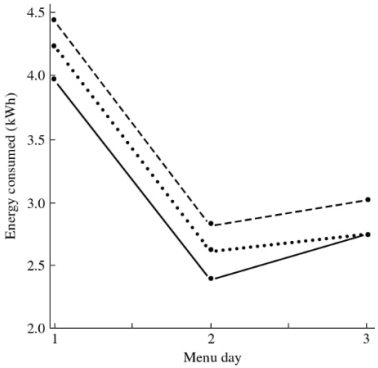


Fig. 12.3. [MSDA] Plots of average energy consumption. Solid line: E1; dotted line: E2; dashed line: E3.

- **Interaction between menu days (Factor A) and ranges (Factor B):** If there is no interaction, the effects are additive, and the profile lines should be perfectly parallel.
- However, the gap between the solid and dotted lines narrowing notably at Menu Day 3 indicates a potential interaction.
- We will study ANOVA, a statistical framework to examine whether these main effects and their interaction are statistically significant.

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Normal Theory: The F Test

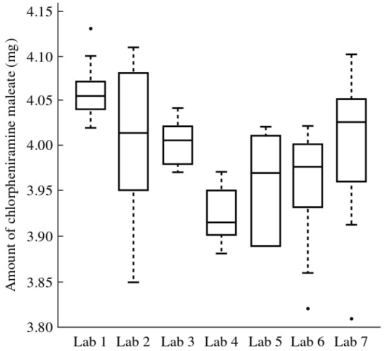


Fig. 12.1. [MSDA] Boxplots of data in Table 12.1.

- The j-th observation from group i is modeled as:

$$\begin{aligned} Y_{ij} &= \mathrm{E}[\bar{Y}_{i\cdot}] + \epsilon_{ij} \\ &= \underbrace{\mathrm{E}[\bar{Y}_{\cdot\cdot}]}_{=:\mu} + \underbrace{\left(\mathrm{E}[\bar{Y}_{i\cdot}] - \mathrm{E}[\bar{Y}_{\cdot\cdot}]\right)}_{=:\alpha_i} + \epsilon_{ij}, \end{aligned} \tag{3}$$

  where μ represents the overall mean and α_i represents the deviation of group i from μ.
- Here, $\bar{Y}_{i\cdot} = n_i^{-1} \sum_{j=1}^{n_i} Y_{ij}$ and $\bar{Y}_{\cdot\cdot} = \left(\sum_{i=1}^{I} n_i\right)^{-1} \sum_{i=1}^{I} \sum_{j=1}^{n_i} Y_{ij}$.

Normal Theory: The F Test

- Under the Gaussian assumption for the measurement errors ϵ_{ij}, the probabilistic model can be expressed as:

$$\begin{cases} Y_{ij} = \mu + \alpha_i + \epsilon_{ij} \\ \sum_{i=1}^{I} n_i \alpha_i = 0 \\ \epsilon_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \end{cases} \tag{4}$$

  where $\theta = (\mu, \alpha_1, \ldots, \alpha_{I-1}, \sigma^2)^T$ parametrizes the model.
- Here, the second line holds because

$$\sum_{i=1}^{I} n_i \left(\bar{Y}_{i\cdot} - \bar{Y}_{\cdot\cdot} \right) = \left(\sum_{i=1}^{I} n_i \bar{Y}_{i\cdot} \right) - \left(\sum_{i=1}^{I} n_i \right) \bar{Y}_{\cdot\cdot} = 0. \tag{5}$$

Normal Theory: The F Test

- Suppose we test:

$$H_0 : \text{There is no treatment effect} \quad \text{vs.} \quad H_1 : \text{Not } H_0. \tag{6}$$

Based on the previous formulations, H_0 can be rewritten as:

$$H_0 : \alpha_1 = \alpha_2 = \cdots = \alpha_I = 0. \tag{7}$$

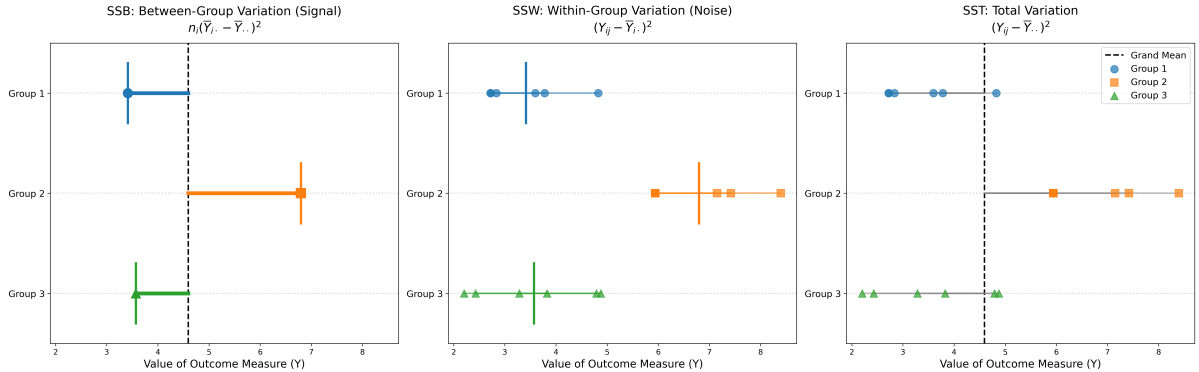
- To understand the sources of variation, we define Sum of Squares (SS):

$$\text{SST} = \sum_{i=1}^{I} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{..})^2 \qquad \text{(Total SS)}$$

$$\text{SSB} = \sum_{i=1}^{I} n_i (\bar{Y}_{i.} - \bar{Y}_{..})^2 \qquad \text{(Between SS)}$$

$$\text{SSW} = \sum_{i=1}^{I} \sum_{j=1}^{n_i} (Y_{ij} - \bar{Y}_{i.})^2 \qquad \text{(Within SS)}$$

Normal Theory: The F Test



Normal Theory: The F Test

Exercise 1

Consider the following one-way ANOVA model:

$$\begin{cases} Y_{ij} = \mu + \alpha_i + \epsilon_{ij} \\ \sum_{i=1}^{I} n_i \alpha_i = 0 \\ \epsilon_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \end{cases} \tag{8}$$

where $\theta = (\mu, \alpha_1, \ldots, \alpha_{I-1}, \sigma^2)^T$ is unknown. Suppose we test:

$$H_0 : \alpha_1 = \cdots = \alpha_I = 0 \quad \text{vs.} \quad H_1 : \text{Not } H_0. \tag{9}$$

Normal Theory: The F Test

a) Show that the following identity holds:

$$\underbrace{\sum_{i=1}^{I}\sum_{j=1}^{n_i}(Y_{ij}-\bar{Y}_{..})^2}_{SST} = \underbrace{\sum_{i=1}^{I} n_i(\bar{Y}_{i.}-\bar{Y}_{..})^2}_{SSB} + \underbrace{\sum_{i=1}^{I}\sum_{j=1}^{n_i}(Y_{ij}-\bar{Y}_{i.})^2}_{SSW}. \qquad (10)$$

Normal Theory: The F Test

b) Show that SSB $\perp\!\!\!\perp$ SSW.

Normal Theory: The F Test

Normal Theory: The F Test

c) Show that, under H_0, $\text{SSB}/\sigma^2 \sim \chi^2(I-1)$ and $\text{SSW}/\sigma^2 \sim \chi^2(N-I)$.

Normal Theory: The F Test

Normal Theory: The F Test

- The likelihood function can be expressed as:

$$L(\theta; \boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_I) = \prod_{i=1}^{I} \prod_{j=1}^{n_i} \left(2\pi\sigma^2\right)^{-1/2} \exp\left(-\frac{(Y_{ij} - \mu - \alpha_i)^2}{2\sigma^2}\right). \tag{11}$$

- Then, the GLR test statistic is given by:

$$-2\log\Lambda(\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_I) = N\log\left(1 + \frac{\text{SSB}}{\text{SSW}}\right), \tag{12}$$

  where $N = \sum_{i=1}^{I} n_i$.
- Therefore, the GLR test of level α is given by:

$$d^{\text{GLR}}(\boldsymbol{Y}_1, \ldots, \boldsymbol{Y}_I) = \begin{cases} 1 & \text{if } W > F_\alpha(I-1, N-I) \\ 0 & \text{if } W \leq F_\alpha(I-1, N-I). \end{cases} \tag{13}$$

Normal Theory: The F Test

Exercise 2

Consider the following one-way ANOVA model:

$$\begin{cases} Y_{ij} = \mu + \alpha_i + \epsilon_{ij} \\ \sum_{i=1}^{I} n_i \alpha_i = 0 \\ \epsilon_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \end{cases} \tag{14}$$

where $\theta = (\mu, \alpha_1, \ldots, \alpha_{I-1}, \sigma^2)^T$ is unknown. Suppose we test:

$$H_0 : \alpha_1 = \cdots = \alpha_I = 0 \quad \text{vs.} \quad H_1 : \text{Not } H_0. \tag{15}$$

Normal Theory: The F Test

a) Show that the MLE over $\Omega = \Omega_0 \cup \Omega_1$ is given by:

$$\hat{\mu}^{\Omega} = \bar{Y}_{..}, \quad \hat{\alpha}_i^{\Omega} = \bar{Y}_{i.} - \bar{Y}_{..}, \quad \text{and} \quad \widehat{\sigma^2}^{\Omega} = \text{SSW}/N, \tag{16}$$

where $N = \sum_{i=1}^{I} n_i$.

Normal Theory: The F Test

Normal Theory: The F Test

b) Show that the MLE over Ω_0 is given by:

$$\hat{\mu}^{\Omega_0} = \bar{Y}_{\cdot\cdot}, \quad \hat{\alpha}_i^{\Omega_0} = 0, \quad \text{and} \quad \widehat{\sigma^2}^{\Omega_0} = \mathrm{SST}/N. \tag{17}$$

Normal Theory: The F Test

Normal Theory: The F Test

c) Show that the GLR test statistic is given by:

$$-2\log\Lambda(\boldsymbol{Y}_1,\ldots,\boldsymbol{Y}_I) = N\log\left(1+\frac{\text{SSB}}{\text{SSW}}\right). \tag{18}$$

Normal Theory: The F Test

Normal Theory: The F Test

- The results of the F-test and the decomposition of variance are conventionally summarized in an Analysis of Variance (ANOVA) Table.
- Let $N = \sum_{i=1}^{I} n_i$ be the total number of observations. The standard structure is:

Source	df	SS	MS	F-statistic
Between (Treatments)	$I-1$	SSB	$\text{MSB} = \frac{\text{SSB}}{I-1}$	$W = \frac{\text{MSB}}{\text{MSW}}$
Within (Error)	$N-I$	SSW	$\text{MSW} = \frac{\text{SSW}}{N-I}$	
Total	$N-1$	SST		

- We reject H_0 at significance level α if the computed F-statistic $W > F_\alpha(I-1, N-I)$.

Normal Theory: The F Test

- Let $\pi_{d^{\text{GLR}}}(\theta)$ be the power function of the GLR test.
- <u>**Non-central** χ^2</u>: If $Z_1, \ldots, Z_k \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,1)$, then $\sum_{i=1}^k (Z_i + \mu_i)^2 \sim \chi^2(k, \lambda)$ where $\lambda = \sum_{i=1}^k \mu_i^2$.
- <u>**Non-central** F</u>: If $V_1 \sim \chi^2(\nu_1, \lambda)$, $V_2 \sim \chi^2(\nu_2)$, and $V_1 \perp\!\!\!\perp V_2$, then $W = \frac{V_1/\nu_1}{V_2/\nu_2} \sim F(\nu_1, \nu_2, \lambda)$.
- Under H_1, SSB$/\sigma^2$ is known to follow a non-central χ^2 distribution:[1]

$$\frac{\text{SSB}}{\sigma^2} \sim \chi^2(I-1, \lambda) \tag{19}$$

  where the non-centrality parameter is $\lambda = \frac{1}{\sigma^2} \sum_{i=1}^I n_i \alpha_i^2$.
- Based on this, the power function can be expressed as $\pi_{d^{\text{GLR}}}(\theta) = P_\theta\left(W > F_\alpha(I-1, N-I)\right)$ where $W \sim F(I-1, N-I, \lambda)$.

[1] The detailed proof is beyond the scope of this course.

The Problem of Multiple Comparisons

- While one-way ANOVA tests $H_0 : \alpha_1 = \cdots = \alpha_I$, a rejection of H_0 does not specify which particular groups differ.
- Furthermore, as I increases, H_0 becomes a "weaker" (more restrictive) hypothesis, making it more likely to be rejected even if only a single pair of groups differs.
- A naive remedy would be testing:

$$H_0^{i,i'} : \alpha_i = \alpha_{i'} \quad \text{vs.} \quad H_1^{i,i'} : \alpha_i \neq \alpha_{i'} \tag{20}$$

for all $1 \leq i < i' \leq I$, totaling $m = \binom{I}{2}$ pairwise t-tests.

The Problem of Multiple Comparisons

- The *Bonferroni method* is a procedure to address the problem of multiple comparisons.
- To control the FWER at level α, it conducts each of the m individual tests at a more stringent level, $\alpha' = \alpha/m$.
- **Bonferroni Inequality**: Let E_i be the event of committing a Type I error for the i-th test. The FWER is then bounded as follows:

$$\text{FWER} = P\left(\bigcup_{i=1}^{m} E_i\right) \leq \sum_{i=1}^{m} P(E_i) = \sum_{i=1}^{m} \frac{\alpha}{m} = \alpha. \tag{21}$$

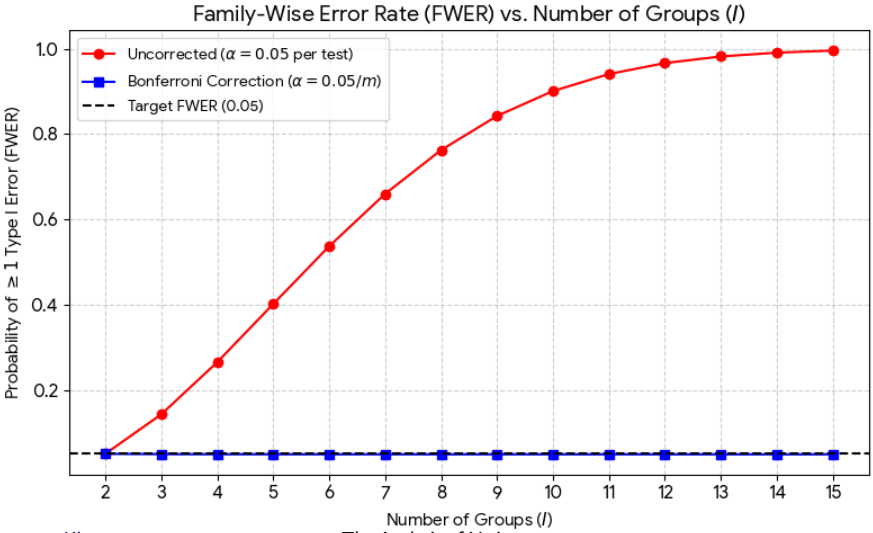
The Problem of Multiple Comparisons

- Based on this, we construct the $1-\alpha$ simultaneous confidence interval for the difference $\alpha_i - \alpha_{i'}$ as follows:

$$(\bar{Y}_{i\cdot\cdot} - \bar{Y}_{i'\cdot\cdot}) \pm t_{1-\frac{\alpha}{2m},\, n_i+n_{i'}-2}\sqrt{S^2_{p,i,i'}\left(\frac{1}{n_i}+\frac{1}{n_{i'}}\right)} \tag{22}$$

- Note that this structure is identical to the standard two independent samples t-test using the pooled variance $S^2_{p,i,i'}$, except the critical value is adjusted to $t_{1-\frac{\alpha}{2m}}$ to account for the m comparisons.
- We reject $H_0^{i,i'} : \alpha_i = \alpha_{i'}$ if 0 does not fall within this interval.

The Problem of Multiple Comparisons



The Problem of Multiple Comparisons

- While the Bonferroni method is simple and the FWER is well-controlled, it becomes overly conservative, leading to low statistical power, especially when m is large or the tests are highly correlated.
- Furthermore, the test assumes homogeneous Gaussian errors, which may be a strong assumption in practice.
- Other popular approaches include the parametric Tukey's method and the non-parametric Kruskal-Wallis test. See Section 12.2.2 for details.

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Normal Theory: The Two-Way Layout

	Factor B		
Factor A	Level 1	$\cdots$	Level J
Level 1	$Y_{111}, \ldots, Y_{11n_{11}}$	$\cdots$	$Y_{1J1}, \ldots, Y_{1Jn_{1J}}$
$\vdots$	$\vdots$	$\ddots$	$\vdots$
Level I	$Y_{I11}, \ldots, Y_{I1n_{I1}}$	$\cdots$	$Y_{IJ1}, \ldots, Y_{IJn_{IJ}}$

- The two-way layout introduces another factor:

$$Y_{ijk} = \mathrm{E}[Y_{ijk}] + \epsilon_{ijk} = \underbrace{\mathrm{E}[\bar{Y}_{\cdots}]}_{=:\mu} + \underbrace{\Big(\mathrm{E}[\bar{Y}_{i\cdot\cdot}] - \mathrm{E}[\bar{Y}_{\cdots}]\Big)}_{=:\alpha_i} + \underbrace{\Big(\mathrm{E}[\bar{Y}_{\cdot j\cdot}] - \mathrm{E}[\bar{Y}_{\cdots}]\Big)}_{=:\beta_j}$$
$$+ \underbrace{\Big(\mathrm{E}[\bar{Y}_{ij\cdot}] - \mathrm{E}[\bar{Y}_{i\cdot\cdot}] - \mathrm{E}[\bar{Y}_{\cdot j\cdot}] + \mathrm{E}[\bar{Y}_{\cdots}]\Big)}_{=:\delta_{ij}} + \epsilon_{ijk}, \tag{23}$$

where δ_{ij} represents the interaction effect between Factors A and B.

Normal Theory for the Two-Way Layout

- In this course, we exclusively consider the *balanced design*, where the number of observations in each cell is a constant $K \geq 2$ (i.e., $n_{ij} = K$ for all i, j).
- Under this balanced assumption, the sample means are defined as:

$$\bar{Y}_{ij\cdot} = \frac{1}{K}\sum_{k=1}^{K} Y_{ijk}, \qquad \bar{Y}_{i\cdot\cdot} = \frac{1}{JK}\sum_{j=1}^{J}\sum_{k=1}^{K} Y_{ijk},$$

$$\bar{Y}_{\cdot j\cdot} = \frac{1}{IK}\sum_{i=1}^{I}\sum_{k=1}^{K} Y_{ijk}, \qquad \bar{Y}_{\cdots} = \frac{1}{N}\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K} Y_{ijk},$$

where $N = IJK$ is the total number of observations.

Normal Theory for the Two-Way Layout

- Under the Gaussian assumption for the measurement errors ϵ_{ijk}, the probabilistic model for a balanced design is:

$$\begin{cases} Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \epsilon_{ijk} \\ \sum_{i=1}^{I} \alpha_i = 0, \quad \sum_{j=1}^{J} \beta_j = 0 \\ \sum_{i=1}^{I} \delta_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^{J} \delta_{ij} = 0 \text{ for all } i \\ \epsilon_{ijk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2) \end{cases} \tag{24}$$

  for $i = 1, \ldots, I$, $j = 1, \ldots, J$, and $k = 1, \ldots, K$.
- Here, $\theta = \left(\mu, \alpha_1, \ldots, \alpha_{I-1}, \beta_1, \ldots, \beta_{J-1}, \delta_{11}, \ldots, \delta_{I-1,J-1}, \sigma^2\right)^T$ parametrizes the model.

Normal Theory for the Two-Way Layout

- In a two-way ANOVA, we typically test three distinct hypotheses:
  1. Test for Interaction Effects:

$$H_0^{AB} : \delta_{ij} = 0 \text{ for all } i, j \quad \text{vs.} \quad H_1^{AB} : \text{Not } H_0^{AB} \text{ (at least one } \delta_{ij} \neq 0) \tag{25}$$

  2. Test for Main Effect of Factor A:

$$H_0^{A} : \alpha_1 = \cdots = \alpha_I = 0 \quad \text{vs.} \quad H_1^{A} : \text{Not } H_0^{A} \tag{26}$$

  3. Test for Main Effect of Factor B:

$$H_0^{B} : \beta_1 = \cdots = \beta_J = 0 \quad \text{vs.} \quad H_1^{B} : \text{Not } H_0^{B} \tag{27}$$

- In particular, when there is a significant interaction effect, it is often meaningless to test or interpret the marginal main effects.
- This is because the effect of one factor is not constant across the levels of the other factor, making any overall average across groups highly misleading.

Normal Theory for the Two-Way Layout

- To test these hypotheses, we decompose the SST, $\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}(Y_{ijk}-\bar{Y}_{\cdots})^2$, into four orthogonal components:

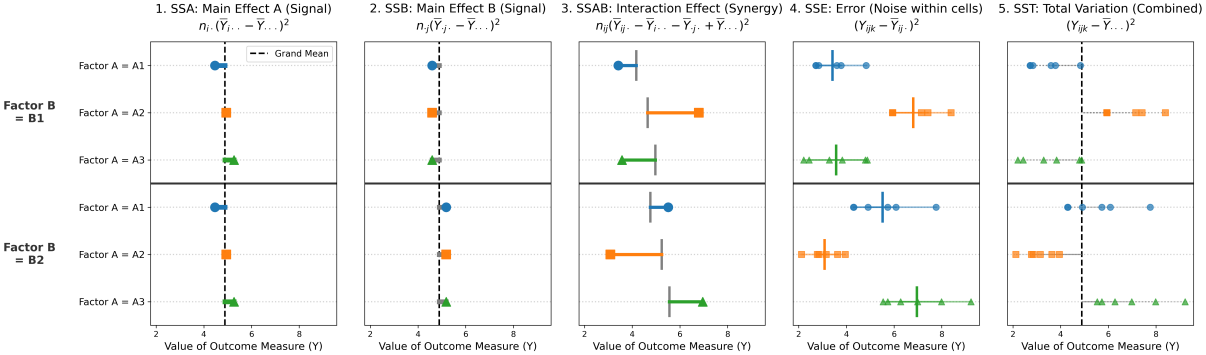
$$\text{SSA} = JK\sum_{i=1}^{I}(\bar{Y}_{i\cdot\cdot}-\bar{Y}_{\cdots})^2 \qquad \text{(Factor A SS)}$$

$$\text{SSB} = IK\sum_{j=1}^{J}(\bar{Y}_{\cdot j\cdot}-\bar{Y}_{\cdots})^2 \qquad \text{(Factor B SS)}$$

$$\text{SSAB} = K\sum_{i=1}^{I}\sum_{j=1}^{J}(\bar{Y}_{ij\cdot}-\bar{Y}_{i\cdot\cdot}-\bar{Y}_{\cdot j\cdot}+\bar{Y}_{\cdots})^2 \qquad \text{(Interaction SS)}$$

$$\text{SSE} = \sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}(Y_{ijk}-\bar{Y}_{ij\cdot})^2 \qquad \text{(Error SS)}$$

Normal Theory for the Two-Way Layout



Two-Way ANOVA: Interaction Effect

Exercise 3

We consider the following balanced two-way ANOVA model:

$$\begin{cases} Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \epsilon_{ijk} \text{ where } 1 \le i \le I,\ 1 \le j \le J, \text{ and } 1 \le k \le K \\ \sum_{i=1}^{I} \alpha_i = 0, \quad \sum_{j=1}^{J} \beta_j = 0, \quad \sum_{i=1}^{I} \delta_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^{J} \delta_{ij} = 0 \text{ for all } i \\ \epsilon_{ijk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \end{cases}$$

where $\theta = \left(\mu, \alpha_1, \ldots, \alpha_{I-1}, \beta_1, \ldots, \beta_{J-1}, \delta_{11}, \ldots, \delta_{I-1,J-1}, \sigma^2\right)^T$ is unknown. Suppose we test $H_0^{AB} : \delta_{ij} = 0$ for all i, j vs. H_1^{AB} : Not H_0^{AB}.

Two-Way ANOVA: Interaction Effect

a) Show that the following identity holds:

$$\underbrace{\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}(Y_{ijk}-\bar{Y}_{\cdots})^2}_{\text{SST}} = \underbrace{JK\sum_{i=1}^{I}(\bar{Y}_{i\cdot\cdot}-\bar{Y}_{\cdots})^2}_{\text{SSA}} + \underbrace{IK\sum_{j=1}^{J}(\bar{Y}_{\cdot j\cdot}-\bar{Y}_{\cdots})^2}_{\text{SSB}}$$
$$+ \underbrace{K\sum_{i=1}^{I}\sum_{j=1}^{J}(\bar{Y}_{ij\cdot}-\bar{Y}_{i\cdot\cdot}-\bar{Y}_{\cdot j\cdot}+\bar{Y}_{\cdots})^2}_{\text{SSAB}}$$
$$+ \underbrace{\sum_{i=1}^{I}\sum_{j=1}^{J}\sum_{k=1}^{K}(Y_{ijk}-\bar{Y}_{ij\cdot})^2}_{\text{SSE}}. \tag{28}$$

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

b) Show that SSAB $\perp\!\!\!\perp$ SSE.

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

c) Show that, under H_0^{AB}, $\text{SSAB}/\sigma^2 \sim \chi^2((I-1)(J-1))$ and $\text{SSE}/\sigma^2 \sim \chi^2(IJ(K-1))$.

Two-Way ANOVA: Interaction Effect

- The likelihood function can be expressed as:

$$L(\theta; \boldsymbol{Y}_{11}, \ldots, \boldsymbol{Y}_{IJ}) = \prod_{i=1}^{I}\prod_{j=1}^{J}\prod_{k=1}^{K}\left(2\pi\sigma^2\right)^{-1/2}\exp\left(-\frac{(Y_{ijk}-\mu-\alpha_i-\beta_j-\delta_{ij})^2}{2\sigma^2}\right). \tag{29}$$

- Then, the GLR test statistic is given by:

$$-2\log\Lambda_{AB} = N\log\left(1+\frac{\text{SSAB}}{\text{SSE}}\right). \tag{30}$$

- Therefore, the GLR test of level α is given by:

$$d_{AB}^{\text{GLR}} = \begin{cases} 1 & \text{if } W_{AB} > F_\alpha((I-1)(J-1), IJ(K-1)) \\ 0 & \text{if } W_{AB} \leq F_\alpha((I-1)(J-1), IJ(K-1)) \end{cases} \tag{31}$$

where $W_{AB} = \frac{\text{SSAB}/((I-1)(J-1))}{\text{SSE}/(IJ(K-1))}$.

Two-Way ANOVA: Interaction Effect

Exercise 4

We consider the following balanced two-way ANOVA model:

$$\begin{cases} Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \epsilon_{ijk} \text{ where } 1 \le i \le I,\ 1 \le j \le J, \text{ and } 1 \le k \le K \\ \sum_{i=1}^{I} \alpha_i = 0, \quad \sum_{j=1}^{J} \beta_j = 0, \quad \sum_{i=1}^{I} \delta_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^{J} \delta_{ij} = 0 \text{ for all } i \\ \epsilon_{ijk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \end{cases}$$

where $\theta = \left(\mu, \alpha_1, \ldots, \alpha_{I-1}, \beta_1, \ldots, \beta_{J-1}, \delta_{11}, \ldots, \delta_{I-1,J-1}, \sigma^2\right)^T$ is unknown. Suppose we test $H_0^{AB} : \delta_{ij} = 0$ for all i, j vs. H_1^{AB} : Not H_0^{AB}.

Two-Way ANOVA: Interaction Effect

a) Show that the MLE over $\Omega = \Omega_0 \cup \Omega_1$ is given by:

$$\begin{aligned} &\hat{\mu}^{\Omega} = \bar{Y}_{\cdots}, \quad \hat{\alpha}_i^{\Omega} = \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdots}, \quad \hat{\beta}_j^{\Omega} = \bar{Y}_{\cdot j\cdot} - \bar{Y}_{\cdots}, \\ &\hat{\delta}_{ij}^{\Omega} = \bar{Y}_{ij\cdot} - \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdot j\cdot} + \bar{Y}_{\cdots}, \quad \text{and} \quad \widehat{\sigma^2}^{\Omega} = \frac{\text{SSE}}{N}. \end{aligned} \tag{32}$$

where $N = IJK$.

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

b) Show that the MLE over Ω_0 (where $\delta_{ij} = 0$) is given by:

$$\hat{\mu}^{\Omega_0} = \bar{Y}_{\cdots}, \quad \hat{\alpha}_i^{\Omega_0} = \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdots}, \quad \hat{\beta}_j^{\Omega_0} = \bar{Y}_{\cdot j\cdot} - \bar{Y}_{\cdots},$$
$$\hat{\delta}_{ij}^{\Omega_0} = 0, \quad \text{and} \quad \widehat{\sigma^2}^{\Omega_0} = \frac{\text{SSAB} + \text{SSE}}{N}, \tag{33}$$

where $N = IJK$.

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

Two-Way ANOVA: Interaction Effect

c) Show that the GLR test statistic is given by:

$$-2\log\Lambda_{AB} = N\log\left(1+\frac{\text{SSAB}}{\text{SSE}}\right). \tag{34}$$

Two-Way ANOVA: Interaction Effect

- Let $N = IJK$ be the total number of observations. The Two-Way ANOVA table is constructed as follows:

Source	df	SS	MS	F-statistic
Factor A	$I-1$	SSA	$\text{MSA} = \frac{\text{SSA}}{I-1}$	$W_A = \frac{\text{MSA}}{\text{MSE}}$
Factor B	$J-1$	SSB	$\text{MSB} = \frac{\text{SSB}}{J-1}$	$W_B = \frac{\text{MSB}}{\text{MSE}}$
Interaction (AB)	$(I-1)(J-1)$	SSAB	$\text{MSAB} = \frac{\text{SSAB}}{(I-1)(J-1)}$	$W_{AB} = \frac{\text{MSAB}}{\text{MSE}}$
Error	$IJ(K-1)$	SSE	$\text{MSE} = \frac{\text{SSE}}{IJ(K-1)}$	
Total	$N-1$	SST		

- We reject $H_0^{AB} : \delta_{ij} = 0$ for all i, j at significance level α if $W_{AB} > F_\alpha((I-1)(J-1), IJ(K-1))$.

Two-Way ANOVA: Main Effect

- If we fail to reject H_0^{AB}, we can proceed to analyze the main effects.
- The GLR test statistics for

$$H_0^A : \alpha_1 = \cdots = \alpha_I = 0 \quad \text{vs.} \quad H_1^A : \text{Not } H_0^A \tag{35}$$

and

$$H_0^B : \beta_1 = \cdots = \beta_J = 0 \quad \text{vs.} \quad H_1^B : \text{Not } H_0^B \tag{36}$$

are given by:

$$-2 \log \Lambda_A = N \log \left(1 + \frac{\text{SSA}}{\text{SSE}}\right) \quad \text{and} \quad -2 \log \Lambda_B = N \log \left(1 + \frac{\text{SSB}}{\text{SSE}}\right). \tag{37}$$

Two-Way ANOVA: Main Effect

- Therefore, the GLR test of level α for testing H_0^A vs. H_1^A is given by:

$$d_A^{\text{GLR}} = \begin{cases} 1 & \text{if } W_A > F_\alpha(I-1, IJ(K-1)) \\ 0 & \text{if } W_A \leq F_\alpha(I-1, IJ(K-1)) \end{cases} \tag{38}$$

  where $W_A = \frac{\text{SSA}/(I-1)}{\text{SSE}/(IJ(K-1))}$.

- Similarly, the GLR test for H_0^B vs. H_1^B is given by:

$$d_B^{\text{GLR}} = \begin{cases} 1 & \text{if } W_B > F_\alpha(J-1, IJ(K-1)) \\ 0 & \text{if } W_B \leq F_\alpha(J-1, IJ(K-1)) \end{cases} \tag{39}$$

  where $W_B = \frac{\text{SSB}/(J-1)}{\text{SSE}/(IJ(K-1))}$.

- The derivation is assigned as a practice problem.

Practice Problem I

1. We consider the following balanced two-way ANOVA model:

$$\begin{cases} Y_{ijk} = \mu + \alpha_i + \beta_j + \delta_{ij} + \epsilon_{ijk} \text{ where } 1 \le i \le I,\ 1 \le j \le J, \text{ and } 1 \le k \le K \\ \sum_{i=1}^{I} \alpha_i = 0, \quad \sum_{j=1}^{J} \beta_j = 0, \quad \sum_{i=1}^{I} \delta_{ij} = 0 \text{ for all } j, \quad \sum_{j=1}^{J} \delta_{ij} = 0 \text{ for all } i \\ \epsilon_{ijk} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \end{cases}$$

where $\theta = \left(\mu, \alpha_1, \ldots, \alpha_{I-1}, \beta_1, \ldots, \beta_{J-1}, \delta_{11}, \ldots, \delta_{I-1,J-1}, \sigma^2\right)^T$ is unknown. Suppose we test:

$$H_0^A : \alpha_1 = \cdots = \alpha_I = 0 \quad \text{vs.} \quad H_1^A : \text{Not } H_0^A. \tag{40}$$

Practice Problem II

a) Show that SSA $\perp\!\!\!\perp$ SSE.
b) Show that, under H_0^A, $\text{SSA}/\sigma^2 \sim \chi^2(I-1)$ and $\text{SSE}/\sigma^2 \sim \chi^2(IJ(K-1))$.
c) The MLE over $\Omega = \Omega_0 \cup \Omega_1$ is given by:

$$\begin{aligned} &\hat{\mu}^{\Omega} = \bar{Y}_{\cdots}, \quad \hat{\alpha}_i^{\Omega} = \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdots}, \quad \hat{\beta}_j^{\Omega} = \bar{Y}_{\cdot j\cdot} - \bar{Y}_{\cdots}, \\ &\hat{\delta}_{ij}^{\Omega} = \bar{Y}_{ij\cdot} - \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdot j\cdot} + \bar{Y}_{\cdots}, \quad \text{and} \quad \widehat{\sigma^2}^{\Omega} = \frac{\text{SSE}}{N}, \end{aligned} \tag{41}$$

where $N = IJK$. Show that the MLE over Ω_0 (where $\alpha_i = 0$) is given by:

$$\begin{aligned} &\hat{\mu}^{\Omega_0} = \bar{Y}_{\cdots}, \quad \hat{\alpha}_i^{\Omega_0} = 0, \quad \hat{\beta}_j^{\Omega_0} = \bar{Y}_{\cdot j\cdot} - \bar{Y}_{\cdots}, \\ &\hat{\delta}_{ij}^{\Omega_0} = \bar{Y}_{ij\cdot} - \bar{Y}_{i\cdot\cdot} - \bar{Y}_{\cdot j\cdot} + \bar{Y}_{\cdots}, \quad \text{and} \quad \widehat{\sigma^2}^{\Omega_0} = \frac{\text{SSA} + \text{SSE}}{N}. \end{aligned} \tag{42}$$

Practice Problem III

d) Show that the GLR test statistic is given by:

$$-2\log\Lambda_A = N\log\left(1+\frac{\text{SSA}}{\text{SSE}}\right). \tag{43}$$

References I

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